Trust Calibration and Collaborative Decision-Making in Human–Agent Sequential Graph Colouring: A Within-Subjects Experimental Study

Abstract

We present a controlled experiment investigating how humans calibrate trust in AI agents across differing levels of agent autonomy and plan quality. The task is a sequential graph-colouring problem in which a human participant and two rule-based AI agents jointly colour the nodes of a graph, earning individual and shared scores that depend on colour choices. The payoff structure creates a deliberate tension: each participant's individually preferred colour differs from the shared-optimal colour, so maximising the team score requires coordination and a degree of individual sacrifice. Three experimental conditions vary the degree of agent autonomy and plan quality: in Mode 1, agents issue per-node proposals with explicit rationales and the human decides; in Mode 2A, agents pre-plan the full colouring with 80% probability of picking the shared-optimal colour; in Mode 2B, the same autonomous planning process is used but with only a 30% probability of choosing the shared-optimal colour. Critically, the interface for Mode 2A and Mode 2B is visually identical, so participants cannot detect quality differences from presentation alone. We measure shared score, individual score, plan-modification rate, explanation engagement, and decision latency, together with score improvement across repeated attempts. The study offers a controlled platform for examining trust calibration, appropriate reliance, and the interaction between agent transparency and agent competence in human–AI collaborative settings.

1 Introduction

A central challenge in human–AI collaboration is calibrating trust appropriately: humans should rely on AI agents when those agents are competent and override them when they are not [**? ?**]. Miscalibrated trust takes two forms: *overtrust*, in which users accept AI recommendations even when the AI is wrong, and *undertrust*, in which users override good recommendations unnecessarily. Both forms are costly, but overtrust receives more attention in safety-critical domains because it can cause humans to defer to confident-looking but incorrect AI outputs.

A key variable shaping trust calibration is the *transparency* of the AI's reasoning process. When an agent shows its rationale—why it recommends a particular action, what information it used, what trade-offs it considered—users have more material with which to evaluate reliability. Autonomous agents that produce pre-made plans and present them for human approval represent a particular transparency challenge: the user can see the recommendation but not the planning process that produced it.

We study these dynamics using a graph-colouring task that operationalises individual-vs-team trade-offs in a simple, reproducible setting. The graph is divided among three participants (one human, two AI agents), each controlling a distinct cluster of nodes. Participants colour their nodes sequentially, one at a time, in a fixed order. The payoff structure is designed so that each participant's selfishly preferred colour differs from the colour that maximises the shared team score. This creates a coordination dilemma: a participant who always chooses their personally preferred colour earns more individually but less as a team; a participant who always chooses the shared-optimal colour sacrifices personal gain for collective benefit.

Three experimental conditions differ in how AI agent advice is delivered:

Mode 1 (Manual Collaborative) For each node in the human's cluster, both AI agents issue a proposed colour with a short written rationale and a table of expected scores. The human sees these proposals simultaneously and then makes their own colour choice, unconstrained by the proposals. Agent nodes are coloured automatically by the respective agent's own rule-based logic.

Mode 2A (Autonomous, High Quality) Before colouring begins, a planning agent runs a deliberation process over the full node sequence and produces a complete proposed colouring. The plan is then presented to the human one step at a time; the human can accept or override each step. The planning process picks the shared-optimal colour with probability 0.8, so roughly 80% of plan steps are objectively good for the team.

Mode 2B (Autonomous, Low Quality) Structurally identical to Mode 2A. The interface, terminology, and presentation are the same. The only difference is that the planning process picks the shared-optimal colour with probability 0.3, producing substantially lower-quality plans. Participants are not told which quality condition they are in.

This design allows us to ask: when the interface is held constant and only the underlying plan quality changes, do participants adjust their override behaviour appropriately? Do participants who engage with agent rationales (by clicking to expand them) show better calibration? Does score improve across repeated attempts (up to three per session) as participants learn the task and the agents' tendencies?

The rest of this paper is organised as follows. Section 2 reviews related work. Section 3 defines the sequential graph-colouring task and its scoring model. Section 4 describes the graph topologies used in the study. Section 5 specifies the three experimental conditions in detail. Section 6 describes the agent models underlying each condition. Section 7 describes the interface. Section 8 describes the experimental procedure. Section 9 defines the dependent variables and hypotheses. Section 10 presents the results (to be completed).

2 Related Work

2.1 Trust in Automation and AI

Lee and See [**?**] define trust in automation as "the attitude that an agent will help achieve an individual's goals in a situation characterised by uncertainty and vulnerability." A large body of literature has documented that trust does not automatically track competence: humans exhibit automation bias (uncritical acceptance of automated outputs) [**?**] as well as automation disuse (ignoring reliable automation). Appropriate reliance is the goal: Buccinca et al. [**?**] show that cognitive forcing functions—interventions that require users to reason before viewing AI recommendations—can reduce overtrust. Our design takes a different approach, varying the quality of the underlying agent while holding the interface constant, to test whether participants detect quality differences behaviourally.

2.2 Human–AI Teaming in Optimisation Tasks

Several studies have used combinatorial optimisation tasks to investigate human–AI collaboration. Wilder et al. [**?**] study human–AI teams on scheduling problems, finding that humans override AI recommendations more when AI explanations are present but not always more accurately. Lai and Tan [**?**] find that providing model explanations improves human decision accuracy only when explanations correctly reflect the model's actual reasoning. Our setting isolates a similar question: does having access to an explanation (by clicking "Explain" in Mode 2) correlate with better override decisions?

2.3 Sequential Decision-Making and Path Dependency

The sequential colouring structure of our task introduces path dependency: a suboptimal choice early in the sequence constrains later options. This mirrors real-world scenarios such as project planning, resource allocation, and shift scheduling, where decisions accumulate. Kahneman et al. [**?**] document how early choices anchor subsequent ones; in our setting, the first few node colour decisions may set an inertial trajectory that the human then accepts rather than revises. The repeated-attempt structure (up to three attempts per session) allows us to measure whether and how quickly participants learn to break this inertia.

2.4 Graph Colouring as a Coordination Task

Graph colouring has been used as a canonical coordination task in distributed constraint optimisation [**? ?**]. Unlike classical DCOP settings, our version does not penalise constraint violations as hard constraints—all colourings are valid, but only some are *optimal* under a shared scoring function. This makes the task an *optimisation under conflicting incentives* rather than a feasibility problem, which is both more realistic (agents' interests often misalign partially rather than completely) and allows continuous measurement of coordination quality via score.

3 Task: Sequential Graph Colouring

3.1 Graph Structure

The experimental task uses an undirected graph $G = (V, E)$ partitioned among three participants: a human h and two AI agents a_1 (AgentA) and a_2 (AgentB). Each participant controls a disjoint cluster of nodes:

$$V = V_h \sqcup V_{a_1} \sqcup V_{a_2}$$

where V_p denotes the set of nodes owned by participant p. Edges E include both *intra-cluster* edges (between nodes in the same cluster) and *inter-cluster* edges (between nodes in different clusters). Inter-cluster edges are the coordination interface: a colour chosen for a node in one cluster may clash with or constrain choices in adjacent clusters.

Each participant assigns a colour from the shared domain $\mathcal{C} = \{\text{red}, \text{blue}, \text{green}\}$ to each of their nodes. Unlike standard graph-colouring problems, there is no *hard* constraint requiring adjacent nodes to have different colours. Instead, adjacency conflicts incur a *score penalty* described in Section 3.3. This design choice ensures that all colourings are valid (the task always terminates) while retaining the incentive to avoid conflicts.

3.2 Sequential Colouring

Nodes are coloured *one at a time* in a fixed predetermined order. The default order is: all human-owned nodes (V_h), followed by all AgentA-owned nodes (V_{a_1}), followed by all AgentB-owned nodes (V_{a_2}). Nodes within each cluster follow the order specified by the graph definition.

This sequential structure introduces *path dependency*: once a node is coloured, its choice constrains (or at minimum influences) the optimal choices for subsequently coloured adjacent nodes. Because human nodes are coloured first, the human's decisions constrain the agents more than the agents' decisions constrain the human. This is an intentional asymmetry: we are interested in how humans make decisions while anticipating (or failing to anticipate) their downstream effects on agent scoring.

An optional *random turn order* mode shuffles the ordering within each attempt. When enabled with the `include_agent_turns` flag, agent nodes may be interspersed with human nodes; when disabled (default), only human nodes are reordered among themselves, and all agent nodes follow. In both cases, the planning agent (Mode 2) re-plans according to the actual shuffled order, ensuring the plan corresponds to the sequence presented to the user.

Participants cannot skip or reorder nodes. They colour nodes strictly in the presented sequence: in Mode 1, they see agent proposals for the current node, then choose; in Mode 2, they see the agent's pre-planned recommendation for the current node, then accept or override.

3.3 Scoring Model

The scoring system supports both individual and collective performance measurement. Each colour assignment earns points for *every* participant simultaneously, with different amounts per participant. Table 1 shows the default per-node point values.

Table 1: Default per-node point values for each colour and participant. Shared total is the sum across all three participants. Each participant's individually preferred colour (maximum personal gain) is shown in bold.

Colour	Human	AgentA	AgentB	Shared Total
Red	**4**	1	2	7
Blue	1	**4**	2	7
Green	2	2	**4**	8

The key feature of this payoff structure is that:

1. Each participant has a *different* individually preferred colour (the colour that maximises their personal points on any given node).
2. The *shared-optimal* colour — the colour that maximises the sum of all participants' points — is green (8 pts total), which is AgentB's preferred colour but not the preferred choice for Human or AgentA.
3. Choosing one's own preferred colour sacrifices 1 point of shared score per node relative to always choosing green.

This creates a *social dilemma* structure embedded within a sequential colouring task. A human who always chooses red earns the maximum personal score (4 pts/node) but contributes only 7 pts/node to the shared pool. A human who defers to green earns only 2 pts/node personally but contributes 8 pts/node to the shared total. Whether participants sacrifice personal gain for shared benefit — and whether agent advice moves them in either direction — is a primary outcome of interest.

Clash Penalty. If two adjacent nodes (sharing an edge in E) are assigned the same colour, a penalty of 10 points is deducted from the shared score. For intra-participant clashes (both nodes owned by the same participant), the full penalty is deducted from that participant's individual score. For inter-participant clashes (adjacent nodes owned by different participants), the penalty is split: $\lfloor 10/2 \rfloor = 5$ points from each owner. Formally, the score for a complete assignment $x : V \to \mathcal{C}$ is:

$$\text{Score}_p(x) = \sum_{v \in V_p} \text{pts}_p(x_v) - \sum_{\substack{(u,v) \in E \\ x_u = x_v}} \text{penalty}(u, v, p) \tag{1}$$

where $\text{pts}_p(c)$ is participant p's point value for colour c (from Table 1), and $\text{penalty}(u, v, p)$ is the portion of the clash penalty on edge (u, v) allocated to participant p.

The shared score is $\text{Score}_{shared}(x) = \sum_p \text{Score}_p(x)$. The maximally achievable shared score, with no clashes, is $8 \times |V|$ (all nodes green). The minimum achievable shared score depends on the graph topology and the number of edges.

Score interpretation. Because the graph is fixed within a session, the shared score is bounded above and below by constants determined by the topology. The key observable is therefore the *fraction of the maximum shared score* achieved, and the *deviation from optimal* (all-green assignment). A secondary observable is the *individual-vs-team trade-off*: participants who choose red more often earn more personally but contribute less collectively.

3.4 Repeated Attempts

Each experimental session consists of up to A attempts (default $A = 3$) on the same graph instance. After completing an attempt, the participant sees a results screen showing the final colouring, a score breakdown, and (from the second attempt onwards) a comparison with previous attempts. They may then choose to start another attempt or end the session.

Each new attempt begins with the graph in the same initial (uncoloured) state; the previous attempt's colouring is displayed as a visual reference but does not constrain the new attempt. In Mode 2, a new plan is generated for each attempt using the same random seed incremented by the attempt number, so the plan for attempt k is a deterministic function of the graph, the seed, and k. Across attempts, planning quality is held constant (same mode, same k-dependent seed), but the sequential nature of the task means participants who learned from the previous attempt can apply that knowledge.

4 Graph Topologies

The study includes a family of systematically designed graph instances spanning three cluster sizes and two topological families. All graphs are defined with nodes partitioned into two or three agent clusters plus one human cluster, and all use the same colour domain $\mathcal{C} = \{\text{red}, \text{blue}, \text{green}\}$.

4.1 Standard Cluster Graphs

Standard graphs are characterised by dense intra-cluster connectivity and sparse inter-cluster edges. Table 2 summarises the variants.

Table 2: Standard cluster graph variants. n = total nodes; n_c = nodes per cluster; $|E|$ = total edges.

Name	**Description**	n	n_c	$\|E\|$
`s9_3c`	3 clusters $\times$ 3 nodes; internal triangles; H–A and H–B cross-edges	9	3	12
`s9_3c_xab`	`s9_3c` + explicit A–B cross-links (2 edges)	9	3	14
`s12_3c`	3 clusters $\times$ 4 nodes; quad + diagonal per cluster; 6 cross-edges	12	4	17
`s12_3c_xab`	`s12_3c` + A–B cross-links (2 edges)	12	4	19
`s18_3c`	3 clusters $\times$ 6 nodes; 6-cycle + 2 chords; 6 cross-edges	18	6	26
`s18_3c_xab`	`s18_3c` + A–B cross-links (3 edges)	18	6	29
`s9_3c_c`	4-player variant of `s9_3c` (adds cluster)	12	3	–
`s12_3c_c`	4-player variant of `s12_3c`	16	4	–
`s18_3c_c`	4-player variant of `s18_3c`	24	6	–

The `xab` variants add direct cross-edges between the two agent clusters (not mediated by the human cluster), increasing inter-agent coupling and reducing the human's structural centrality in the coordination problem.

4.2 Hub Topology Graphs

Hub graphs use a *star topology* within each cluster: one hub node is connected to all satellite nodes in the cluster. This topology concentrates constraint propagation through the hub node, creating a cleaner path-dependency structure. Table 3 summarises the hub variants.

Table 3: Hub topology graph variants. n = total nodes; architecture is the intra-cluster structure; cross-edges is the inter-cluster connection pattern.

Name	Architecture	n	Key cross-edges
`hub12`	Star (hub + 3 satellites) per cluster	12	h_1–a_1, h_1–b_1
`hub12_tri`	`hub12` + a_1–b_1	12	Hub triangle
`hub15_deep`	Hub triangle + h_2–a_1, h_3–b_1	15	Deep cross-constraints
`hub18`	Hub + satellite pairs per cluster; full hub triangle	18	Hub triangle + pair threading
`hub18b`	Asymmetric: H chain, A disjoint pairs, B chain	18	Hub triangle + asymmetric links

Hub topologies have the property that the hub node's colour forces or strongly constrains the colours of adjacent cluster hubs (via inter-cluster edges). In `hub12`, for example, colouring h_1 (the human hub) to red immediately constrains a_1 (the AgentA hub) to be non-red, and b_1 (the AgentB hub) to also be non-red. This cascades through both agent clusters.

The asymmetric `hub18b` variant models realistic organisational heterogeneity: the human cluster and AgentB cluster have chain internal structure (which propagates colour constraints along the chain), while AgentA's cluster has disjoint pairs (which do not propagate constraints as strongly). This creates an interesting scenario where the human's constraints ripple through structurally different agent clusters at different rates.

4.3 Colouring Order

In the default (non-random) turn order, nodes are coloured in a fixed sequence: all human-cluster nodes, then all AgentA-cluster nodes, then all AgentB-cluster nodes. Within each cluster, nodes follow the order defined in the graph specification (typically hub first, then satellites in index order).

This ordering ensures that the human acts first and sets the context for both agents. The agents' automatic colouring decisions (in Mode 1) or plan steps (in Mode 2) are then presented after the human has committed their choices, reflecting the causal structure of the task.

5 Experimental Conditions

The study uses a within-subjects design with three conditions applied to each participant. The conditions are:

Mode 1 — Manual Collaborative Per-node proposals. For each human-owned node, both agents simultaneously issue a proposed colour accompanied by a written rationale. A score-prediction table shows each agent's expected outcome under each possible colour choice. The human reads these proposals and then freely chooses any colour. After all human nodes are coloured, agent nodes are coloured automatically by the agents' own greedy logic.

Mode 2A — Autonomous, High Quality The planning agent generates a complete proposed colouring for all nodes before the session starts, using a quality parameter of $q = 0.8$: each plan step independently picks the shared-optimal colour with probability 0.8, and picks a random safe colour with probability 0.2. This plan is presented to the human one step at a time. For human-owned nodes, the participant can accept the plan's recommendation or override it. For agent-owned nodes, the plan step is applied automatically with a brief visual pause.

Mode 2B — Autonomous, Low Quality Identical in interface and interaction structure to Mode 2A. The planning agent uses quality parameter $q = 0.3$: each plan step picks the shared-optimal colour with probability 0.3 only, making the plan substantially worse on average. Participants are not informed of the quality level; the interface is intentionally indistinguishable from Mode 2A.

Within-subjects counterbalancing. All participants complete all three conditions. Conditions are presented in counterbalanced order across participants to control for learning and fatigue effects. A different graph instance (from the families described in Section 4) is used for each condition. Graph instances are matched on the number of nodes, cluster sizes, and approximate scoring difficulty.

Design rationale. Mode 1 represents a *transparent advisory* interface: the agent is fully explainable at the per-node level and the human always retains final decision authority. Mode 2A represents a *competent autonomous* system: the agent plans well and the human's role shifts toward review and approval. Mode 2B represents an *incompetent autonomous* system that *looks* identical to Mode 2A. The comparison between Mode 2A and Mode 2B is the primary test of whether participants appropriately calibrate trust based on the quality of outcomes rather than the form of the interface. The comparison between Mode 1 and Mode 2 (pooling 2A and 2B) tests the effect of interaction style: per-node deliberation versus pre-planned review.

6 Agent Models

All agent logic is deterministic and rule-based. No large language model or learned model is used. This design choice ensures full reproducibility, eliminates variance due to stochastic model outputs, and allows precise control over agent quality. The term "agent" is used throughout the interface; participants are not told the mechanism.

6.1 Proposal Agent (Mode 1)

The proposal agent (`ProposalAgent`) generates a per-node colour recommendation each time a new human-owned node becomes active. Each AI agent runs this logic independently; both proposals are shown simultaneously.

Adjacency analysis. The agent first identifies which of its own nodes are adjacent (share an edge) to the current human node. If the agent has no adjacent node, it does not generate a proposal for this step. If multiple adjacent nodes exist, the agent selects one reference node using a priority rule: already-coloured adjacent nodes take precedence (actionable advice); among uncoloured nodes, the first in alphabetical order is chosen.

Colour recommendation. The recommendation logic depends on the reference node's status:

- **Reference node already coloured:** Recommend the first colour (in $\{\text{red}, \text{blue}, \text{green}\}$ order) that does not clash with the reference node's colour. This avoids a cross-cluster penalty.
- **Reference node not yet coloured:** The agent anticipates placing its own preferred colour on the reference node later, so it recommends a colour for the human node that avoids clashing with that anticipated choice. Specifically, it recommends the first colour that is *not* the agent's preferred colour.

If no non-clashing colour exists (e.g. all colours are ruled out simultaneously), the agent falls back to the first available colour in domain order.

Rationale generation. Each proposal includes a short rationale (one or two sentences displayed inline) and a full rationale (expandable detail, shown on user request). Short rationales follow one of two templates:

- *Clash avoidance*: "My node X is already [colour] — I recommend [colour] here so our adjacent nodes don't clash."
- *Future planning*: "My node X is next to this one and I plan to colour it [preferred] — so I recommend [colour] here to keep our nodes well-coordinated."

When the attempt number is greater than 1, a note is added comparing the current recommendation to the previous attempt's colour at this node: "Last attempt this was [colour]; my recommendation is [colour] this time."

Full rationales include: the adjacent node identification, the rationale for colour selection, a per-colour point table for all participants, shared-score contributions per colour, and a count of remaining uncoloured nodes.

Expected score display. In addition to the written rationale, Mode 1 shows a score-prediction table (described in Section 7) showing each agent's expected score contribution under each possible colour choice for the current node. This is computed using a forward simulation: for each possible colour c that the human might choose, the agent greedily colours its remaining nodes (preferring clash avoidance, then score maximisation), and reports the agent's resulting total score. This gives participants explicit quantitative information about the consequences of their choices for each agent.

Self-colouring (agent-owned nodes). When an agent-owned node becomes active (after all human nodes are decided), the agent colours it automatically using a greedy policy: avoid clashing with already-coloured adjacent nodes; among clash-free candidates, pick the colour that maximises the agent's own score for that node.

6.2 Planning Agent (Mode 2)

The planning agent (`PlanningAgent`) produces a complete proposed colouring for all nodes *before* the session begins. This plan is then delivered to the participant step by step.

Deliberation process. The planning agent is initialised with a quality parameter $q \in [0, 1]$. In the study, $q = 0.8$ for Mode 2A and $q = 0.3$ for Mode 2B. A seeded random number generator is initialised with seed $s + k$, where s is the session seed and k is the attempt number (1-based), ensuring different but reproducible plans across attempts.

For each node in the colouring sequence, the deliberation process runs as follows:

1. **Clash detection:** Identify all colours already assigned to adjacent nodes (nodes already coloured in the current plan-in-progress). Colours that would clash with an adjacent already-planned colour are

excluded; the remaining colours are the *safe candidates*. If all colours clash (which can occur in dense graphs), all colours are re-admitted as candidates.

2. **Agent votes:** Each sub-agent independently "votes" for its preferred colour (the colour that maximises its own per-node points, regardless of the graph context). Votes are logged in the deliberation record.
3. **Quality-biased selection:** A random draw determines the selection mechanism for this step:
   - With probability q: select the safe candidate with the highest shared-score total (the shared-optimal safe colour). This is the "high-quality" choice.
   - With probability $1 - q$: select uniformly at random from the safe candidates. This may coincidentally be the shared-optimal colour, but is not guided by shared score.
4. **Rationale generation:** A short rationale and full rationale are generated, templated as:
   - Short (shared-optimal): "[Colour] gives the team N pts on this node, which maximises our shared score."
   - Short (random): "[Colour] was selected for [node] after agent deliberation (team score: N pts)."

   The full rationale includes all agent votes, the vote tally per colour, the shared-score contribution of each colour, the selection method, and the final choice.

Deliberation log. The complete deliberation log is accessible to participants via a collapsible panel in the interface (labelled "How was this plan made?"). This log shows the step-by-step deliberation for every node: which colours were safe, how each agent voted, which selection method was used, and what the expected score contribution is. An interaction event is logged whenever a participant opens this panel.

Expected score. The expected shared score for the full plan is computed as the sum of the shared-score contributions of the chosen colour at each node, before considering any clashes. This figure is displayed in the plan summary panel.

Plan invariance. Because the planning process uses a seeded RNG, the same graph instance, seed, and attempt number always produce the same plan. This means Mode 2A and Mode 2B plans are reproducible and their quality is deterministic. Within a condition, different participants see different plans only if they use different seeds (configurable at session launch).

7 Interface

The experiment interface is implemented in Python using the Tkinter GUI toolkit. All conditions use the same application window and layout; only the right-hand panel changes between Mode 1 and Mode 2.

7.1 Window Layout

The main window (1020 $\times$ 660 minimum) is divided into three horizontal zones:

Top — Scoring HUD A horizontal bar showing the current shared score (large, green) and individual score cards for each participant (colour chip + name + points). Updates after every colour decision.

Middle — Main Panel Divided into a left pane (graph display) and a right pane (mode-specific interaction panel).

Bottom — Sequence Bar A breadcrumb row of small cells, one per node, showing colouring progress. Grey cells are uncoloured; filled cells show the assigned colour.

7.2 Graph Display (Left Pane)

The left pane contains two graph canvases stacked vertically:

- **Overview canvas** (440×280 px): Shows the complete graph with all nodes and edges. Nodes are positioned according to precomputed fractional coordinates stored per graph preset. Coloured nodes show their assigned colour; uncoloured nodes are grey.
- **Neighbourhood canvas** (440×210 px): Shows the current active node and its direct neighbours in isolation. Provides a local view that highlights the immediate constraint context.

7.3 Mode 1 Right Panel

The Mode 1 panel contains the following elements from top to bottom:

1. **Reward key**: A small table showing the point values of each colour for the human participant, with the highest-value colour highlighted.
2. **Current node label**: Displays the name of the active node.
3. **Score prediction table**: A grid with rows for each agent that has an adjacent node, and columns for each colour. Each cell shows the agent's predicted score contribution if the human chooses that colour, with a star ($\star$) marking the agent's own preferred colour.
4. **Colour choice buttons**: One button per colour, enabled only after proposals have been shown. Clicking a button records the decision and advances to the next node.

Agent proposals are shown automatically when a human node becomes active. Both agents' proposals and the score table appear simultaneously. When the human clicks a colour button, the decision is logged (including source = "human", decision time, and whether a majority of agents agreed with the choice).

7.4 Mode 2 Right Panel

The Mode 2 panel presents the current plan step:

1. **Proposed colour**: The plan's recommended colour for the current node, displayed prominently.
2. **Short rationale**: One or two sentences from the planning agent's rationale for this recommendation, displayed below the proposed colour.
3. **Action buttons**: Three buttons — *Accept* (records the plan's colour, source = "accepted_plan"), *Explain* (toggles the full rationale text), and *Modify* (opens a sub-panel with colour buttons for overriding the plan, source = "modified_plan").
4. **Full rationale panel**: A scrollable text field (hidden by default, toggled by Explain) showing the complete rationale for this step including all agent votes, vote tallies, and shared-score contributions per colour.
5. **Deliberation log panel**: A collapsible text field accessible via a labelled checkbox, showing the complete planning deliberation log for the whole session.

For agent-owned nodes in Mode 2, the plan step is applied automatically with a 500 ms visual pause, and no buttons are shown. The participant observes the plan being executed but cannot intervene.

7.5 End-of-Attempt Results Panel

When all nodes have been coloured, the colouring panel is replaced in-place by a results view showing:

- A final graph rendering with the completed colouring.
- A score table: each participant's name, preferred colour, individual score, and clash penalty (if any).
- The **shared total** score, displayed prominently.
- A score history table (from the second attempt onwards) listing the shared total for each attempt, with markers indicating the current attempt and the best attempt.

- **Action buttons**: "Start Next Attempt" (if attempts remain) and "Finish Session".

The previous attempt's colouring is preloaded as a visual overlay on the graph at the start of each new attempt (with a 200 ms delay for canvas layout), giving participants a reference without constraining their choices.

7.6 Event Logging

All interactions are written to a JSONL event log at `results/participants/<pid>_<timestamp>/events.jsonl`. Each line is a JSON object with a Unix timestamp and event type. Table 4 lists the key logged events.

Table 4: Key JSONL events logged during each session. All events include a Unix timestamp `ts` and event type `event`.

Event type	Fields logged
`session_start`	participant_id, mode, graph, colours, seed, max_attempts, points_config
`attempt_start`	attempt number (1-based)
`node_ready`	attempt, node, node_index (starts per-node timer)
`proposals_shown`	(Mode 1) attempt, node, proposals [{agent, colour}]
`explanation_click`	(Mode 1) attempt, node, agent name
`colour_chosen`	attempt, node, colour, source, decision_time_s, agents_agreed
`plan_step_shown`	(Mode 2) attempt, node, plan_colour
`plan_accepted`	(Mode 2) attempt, node, plan_colour
`plan_modified`	(Mode 2) attempt, node, plan_colour, chosen_colour
`plan_explanation_clicked`	(Mode 2) attempt, node
`deliberation_log_opened`	(Mode 2) attempt
`attempt_complete`	attempt, assignment {node: colour}, score {total, by_owner}, elapsed_s
`replay_requested`	attempt (user chose to start another attempt)
`session_end`	total_attempts, score_trajectory [int], session_elapsed_s

The `colour_chosen` event includes `source` $\in$ {"human", "accepted_plan", "modified_plan", "agent_auto"}, distinguishing human initiative, plan acceptance, plan override, and automatic agent colouring. The `agents_agreed` field (boolean) records whether a majority of agents whose proposals or plan recommended this node had recommended the chosen colour. Decision time is measured from the moment the node was made active (`node_ready`) to the moment the colour button was pressed.

8 Procedure

8.1 Session Structure

Each session is launched via a graphical configuration screen where the experimenter (or participant, in unattended conditions) enters the participant ID, selects the mode and graph preset, sets the random seed, and specifies the maximum number of attempts. A subprocess then opens the experiment window.

A session consists of the following stages:

1. **Introduction and practice**: Participants are introduced to the graph colouring task, the scoring system, and the interface. (Specific instructions TBD per study protocol.)

2. **Experimental task**: The participant completes the colouring task in the assigned condition, with up to $A = 3$ attempts.
3. **Post-condition questionnaire**: After each condition, participants complete a questionnaire on workload, trust, and perceived agent quality. (Scale details TBD.)

8.2 Within-Attempt Flow

Within each attempt, the interaction proceeds as follows:

1. For Mode 2 only: the planning agent generates the complete plan (logged internally; plan generation is not visible to the participant).
2. The first node in the sequence is activated.
3. For each human-owned node:
   - Mode 1: Agent proposals appear; the participant reads and chooses a colour.
   - Mode 2: The plan's recommendation appears; the participant accepts or overrides.
4. For each agent-owned node: the agent's colour is applied automatically (Mode 1: greedy rule; Mode 2: plan step), with a 500 ms visual pause.
5. When all nodes are coloured, the results panel appears. The score is displayed and compared with previous attempts.
6. The participant chooses "Start Next Attempt" or "Finish Session".

8.3 Randomisation and Reproducibility

All stochastic elements (plan generation, optional node order shuffle) are controlled by a session seed s (default: 42). The plan for attempt k uses RNG seed $s + k$. If random turn order is enabled, the node sequence for attempt k is derived from $s + k$ before plan generation, so plan and presentation order are always consistent. This ensures full reproducibility: given the same participant ID, mode, graph, and seed, the entire session is deterministic.

9 Dependent Variables and Hypotheses

9.1 Primary Outcome: Shared Score

The primary dependent variable is the *shared score* achieved at the end of each attempt. This is the sum of all participants' individual scores after applying clash penalties (Equation 1). The maximum achievable shared score is $8|V|$ (all nodes green, no clashes). We report both raw scores and *efficiency*: $\eta = \text{Score}_{\textit{shared}}/(8|V|)$.

9.2 Secondary Outcomes

Individual vs. shared trade-off. The human participant's individual score relative to the shared score. A participant who consistently chooses red (their preferred colour) earns higher individual scores but lower shared scores.

Plan modification rate (Mode 2 only). The proportion of human-node plan steps that the participant overrides (`source` = "modified_plan"). High modification rate in Mode 2B (low quality) relative to Mode 2A (high quality) is evidence of appropriate trust calibration; similar modification rates across 2A and 2B would suggest participants cannot detect quality differences.

Explanation engagement. The proportion of nodes for which the participant expanded the full rationale (Mode 1: `explanation_click`; Mode 2: `plan_explanation_clicked`). Also, the proportion of attempts in which the deliberation log was opened (`deliberation_log_opened`).

Decision latency. The mean `decision_time_s` per node, computed from event timestamps. Longer decision times may indicate deeper deliberation; shorter times may indicate automatic acceptance.

Score improvement across attempts. The difference in shared score between attempt k and attempt 1, $\Delta_k = \text{Score}^{(k)} - \text{Score}^{(1)}$. Positive Δ indicates learning; negative Δ would be unexpected and may indicate fatigue or strategy abandonment.

Agents-agreed rate. The proportion of decisions for which the majority of agents recommended the colour the participant ultimately chose (`agents_agreed` = True). In Mode 1 this reflects whether the participant followed agent advice; in Mode 2 it is equivalent to the plan-acceptance rate.

9.3 Hypotheses

H1 ***(Quality detection)*** Participants will modify a higher proportion of plan steps in Mode 2B than in Mode 2A, showing appropriate behavioural trust calibration based on perceived plan quality.

H2 ***(Score gap)*** Mode 2A will produce higher shared scores than Mode 2B, and the gap will be larger when modification rates are well-calibrated (i.e., participants override low-quality steps without over-riding high-quality ones).

H3 ***(Transparency benefit)*** Mode 1 will produce better-calibrated decisions than Mode 2, because per-node proposals with explicit rationales give participants more material to evaluate each recommendation individually, compared to a pre-committed plan.

H4 ***(Explanation engagement and calibration)*** Participants who engage with full rationales (explanation click rate above the median) will show better plan modification calibration in Mode 2: they will override low-quality plan steps more often while accepting high-quality steps at similar rates.

H5 ***(Learning across attempts)*** Shared scores will improve significantly from attempt 1 to attempt 3, with the improvement larger in Mode 2B than Mode 2A (because there is more room to improve by over-riding bad recommendations) and larger in Mode 2 than Mode 1 (because learning the agent's plan tendencies is more useful when the same plan structure recurs).

H6 ***(Individual vs. shared sacrifice)*** Participants who accept more agent recommendations (higher agents-agreed rate) will show lower individual scores but higher shared scores, consistent with trading personal gain for collective benefit as the agents advise.

10 Results

This section will be completed once data collection is finished. The subsections below provide the planned analysis structure.

10.1 Overview of Data Collected

[Table: participant count, conditions completed, attempt counts, any exclusions.]

10.2 Primary Outcome: Shared Score by Condition

[ANOVA or mixed-effects model on shared score with condition (Mode 1, 2A, 2B) as the within-subjects factor and attempt number as a covariate. Report means, standard deviations, and effect sizes. Include a figure showing mean shared score per condition across attempts.]

10.3 Plan Modification Rate (Modes 2A vs 2B)

[Paired comparison of modification rates between Mode 2A and Mode 2B. Report whether participants modify more steps in Mode 2B (H1). Break down by node position in sequence to test whether modification rates change as participants progress through an attempt.]

10.4 Explanation Engagement

[Descriptive statistics of explanation click rates per condition. Test H4: do high-engagers show better calibration? Report median split analysis and correlation between engagement and plan modification calibration.]

10.5 Decision Latency

[Mean decision time per node by condition, with standard error. Compare latency for accepted-plan vs. modified-plan steps (Mode 2) and for conditions with vs. without explanation clicks (Mode 1).]

10.6 Score Improvement Across Attempts

[Plot score trajectory (score vs. attempt number) per condition. Test H5 with a linear trend test on score across attempts. Report improvement magnitude $\Delta_3 = \text{Score}^{(3)} - \text{Score}^{(1)}$ by condition.]

10.7 Individual vs. Shared Trade-off

[Scatter plot: individual score vs. shared score per participant per condition. Report correlation between agents-agreed rate and (a) individual score and (b) shared score, testing H6.]

11 Discussion

To be completed. Planned discussion points:

- Whether the interface indistinguishability of Mode 2A and Mode 2B produces overtrust (low modification in 2B) or whether participants can detect quality through outcome patterns across nodes.
- The role of rationale transparency: does Mode 1's per-node explainability produce meaningfully different behaviour from Mode 2's global plan transparency?
- Path dependency effects: are early-sequence decisions predictive of final shared score, and do participants who make globally better early choices (e.g., choosing green for high-adjacency nodes) achieve higher totals?
- Comparison with a naïve baseline (all human nodes red, all agents optimal) to contextualise score achievements.
- Implications for the design of autonomous AI systems: what transparency mechanisms are necessary for users to detect and compensate for poor-quality autonomous plans?

12 Conclusion

To be completed.

We have described a controlled experimental platform for studying trust calibration and collaborative decision-making in sequential human–agent graph colouring. The platform's key features are: (1) a payoff structure that creates an intrinsic tension between individual and collective optimisation; (2) a sequential colouring order that introduces path dependency; (3) three experimental conditions that vary agent autonomy and plan quality while holding the interface constant; and (4) fine-grained event logging that supports analysis of decision timing, explanation engagement, and score trajectories across repeated attempts.

The distinguishing feature of the design is that Mode 2A and Mode 2B are *superficially identical*: same visual layout, same language, same plan-presentation mechanism. Whether participants detect the quality difference and calibrate their override behaviour accordingly is the central empirical question. A finding of overtrust (no calibration) would suggest that interface presentation alone determines reliance patterns, regardless of actual quality. A finding of appropriate calibration would suggest that participants can perform implicit quality inference from outcome patterns within a single session.

A Scoring Model: Formal Specification

A.1 Point Configuration

Let $\mathcal{P} = \{h, a_1, a_2\}$ be the set of participants. The point function $\text{pts} : \mathcal{P} \times \mathcal{C} \to \mathbb{Z}_{\geq 0}$ is given by Table 1 in the main text. The shared-score contribution of colour c (regardless of node or participant) is:

$$\text{shared}(c) = \sum_{p \in \mathcal{P}} \text{pts}(p, c)$$

In the default configuration: $\text{shared(red)} = 7$, $\text{shared(blue)} = 7$, $\text{shared(green)} = 8$.

A.2 Clash Penalty Allocation

For edge $(u, v) \in E$ with $x_u = x_v$ (same colour), the clash penalty of $\delta = 10$ is allocated as follows. Let $\mathrm{owner}(u)$ denote the participant who owns node u.

$$\text{penalty}(u, v, p) = \begin{cases} \delta & \text{if } \mathrm{owner}(u) = \mathrm{owner}(v) = p \\ \lfloor \delta/2 \rfloor + (\delta \bmod 2) & \text{if } \mathrm{owner}(u) = p \neq \mathrm{owner}(v) \\ \lfloor \delta/2 \rfloor & \text{if } \mathrm{owner}(v) = p \neq \mathrm{owner}(u) \\ 0 & \text{otherwise} \end{cases}$$

For $\delta = 10$: intra-participant clashes cost 10 to the owning participant; inter-participant clashes cost 5 to each owner (since $10 \bmod 2 = 0$).

A.3 Optimal and Baseline Scores

For a graph with $|V| = n$ nodes and $|E_\times|$ inter-cluster edges:

- **Upper bound (no clashes, all green)**: $\text{shared(green)} \times n = 8n$.
- **Penalty for all-red**: $\text{shared(red)} \times n = 7n$.
- **Penalty if clashes occur**: each clash deducts 10 from the shared total.

The difference between all-green and all-red is n shared points. For the default 12-node graph, this is 12 shared points, a 12.5% difference on the theoretical maximum.

B Planning Agent: Algorithm

Algorithm 1 gives the pseudocode for the planning agent's deliberation process.

Note that sub-agent votes are recorded in the deliberation log (and shown to participants in the full rationale) but do not affect the colour selection. The selection mechanism depends solely on the quality draw and the shared-score ordering of safe candidates.

C Proposal Agent: Algorithm

Algorithm 2 gives the pseudocode for the proposal agent's per-node recommendation.

D Event Log Schema

Each line of the JSONL log is a JSON object. Below is the complete field specification for the most important events.

Algorithm 1 PlanningAgent.generate_plan(k, prior_attempts, node_order)

Require: Attempt number k, list of prior assignments, node sequence σ, quality $q \in [0, 1]$, seed s

Initialise rng $\leftarrow$ Random$(s + k)$
assignment $\leftarrow \{\}$; plan $\leftarrow$ []
for all v in σ **do**
  adj_colours $\leftarrow \{x_u \mid (u, v) \in E, u \in \text{assignment}\}$
  candidates $\leftarrow \mathcal{C} \setminus$ adj_colours
  if candidates $= \emptyset$ **then**
    candidates $\leftarrow \mathcal{C}$ {admit all if full clash}
  end if
  `// Sub-agent votes (self-interested, ignored for selection)`
  for all agent a **do**
    vote $\leftarrow_c$ pts(a, c)
  end for
  $c^* \leftarrow_{c \in \text{candidates}}$ shared(c) {best safe colour}
  if rng.random() $< q$ **then**
    chosen $\leftarrow c^*$ {quality: pick shared-optimal}
  else
    chosen $\leftarrow$ rng.choice(candidates) {explore randomly}
  end if
  assignment$[v] \leftarrow$ chosen
  Append PlannedStep$(v, \text{chosen}, \ldots)$ to plan
end for
return Plan(plan, $\sum_v$ shared(assignment$[v]$), k)

Algorithm 2 ProposalAgent.propose_for_node(v, assignment_so_far, attempt)

Require: Current node v, partial assignment so far, attempt number

adjacent $\leftarrow \{u \in V_a \mid (u, v) \in E\}$
if adjacent $= \emptyset$ **then**
  return None {no proposal if no adjacent agent node}
end if
coloured_adj $\leftarrow \{u \in \text{adjacent} \mid u \in \text{assignment\_so\_far}\}$
if coloured_adj $\neq \emptyset$ **then**
  ref $\leftarrow$ any $u \in$ coloured_adj
  avoid $\leftarrow$ {assignment_so_far[ref]}
  rationale_type $\leftarrow$ "clash_avoidance"
else
  ref $\leftarrow$ first of adjacent alphabetically
  avoid $\leftarrow$ {preferred_colour(a)}
  rationale_type $\leftarrow$ "future_planning"
end if
candidates $\leftarrow \mathcal{C} \setminus$ avoid
if candidates $= \emptyset$ **then**
  candidates $\leftarrow \mathcal{C}$
end if
recommended $\leftarrow$ candidates[0] {first in domain order}
return NodeProposal$(v, \text{recommended}, \text{rationale}(\text{rationale_type}, \ldots))$

colour_chosen

```
{
  "ts": <float>,              // Unix timestamp
  "event": "colour_chosen",
  "attempt": <int>,           // 1-based
  "node": <str>,              // node identifier (e.g. "H1")
  "colour": <str>,            // "red" | "blue" | "green"
  "source": <str>,            // "human" | "accepted_plan" |
                              //  "modified_plan" | "agent_auto"
  "decision_time_s": <float>, // seconds from node_ready to click
  "agents_agreed": <bool>     // majority of agents recommended this colour
}
```

attempt_complete

```
{
  "ts": <float>,
  "event": "attempt_complete",
  "attempt": <int>,
  "assignment": { <node>: <colour>, ... },
  "score": {
    "total": <int>,
    "by_owner": { <participant>: <int>, ... }
  },
  "elapsed_s": <float>
}
```

session_end

```
{
  "ts": <float>,
  "event": "session_end",
  "total_attempts": <int>,
  "score_trajectory": [<int>, ...],  // shared score per attempt
  "session_elapsed_s": <float>
}
```